

## FIBRONECTIN BINDING PROTEIN AND RHEUMATOID ARTHRITIS

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### ABSTRACT:

Rheumatoid arthritis autoimmune disease, is thought to be caused by an interaction between constitutional and environmental (possibly microbial) factors. *Mycobacterium tuberculosis* is considered one of the causative agent of Rheumatoid arthritis. The Fibronectin-binding protein (FnBp) antigens (30/31) kDa antigen) of *M. tuberculosis* is one of the major secretory molecules and is probably routinely recognized by the host immune system in the early stage of tuberculosis infection. Serum immune complexes, prepared from Rheumatoid arthritis (RA), Tuberculosis (TB) patients and normals, were analyzed for the presence of FnBp by ELISA using CF8 (anti-FnBp monoclonal antibody) and by western blotting using Fibronectin-HRP. Significant differences were noted between normal and patient sera by using the monoclonal antibody CF8 ( $p < 0.001$ ). We have detected a 37 kDa Fibronectin-binding protein in RA patients' serum immune complexes, which are reported to play an important role in bacterial diseases. We have shown that the 30/31 kDa FaBp is a characteristic feature of tuberculosis. This was a quite interesting finding that suggests there are different forms of Fibronectin-binding protein antigens probably involved in pathogenesis of RA.

### INTRODUCTION

Rheumatoid arthritis is a chronic, recurrent, systemic inflammatory disease primarily involving joints. The cause of rheumatoid arthritis is uncertain. It is believed to be an immunological disease, possibly with an autoimmune aetiology.<sup>1</sup> Like most autoimmune diseases, rheumatoid arthritis is thought to be caused by an interaction between constitutional and environmental (possibly microbial) factors.<sup>2</sup> Auto an-

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Antibodies form immune complexes which cause inflammation, localizing predominantly in joints. It is believed that autoimmunity in rheumatoid arthritis is triggered by foreign antigen, usually microbial, in genetically susceptible hosts.<sup>3</sup> Studies performed during the last few decades have suggested a variety of different agents, both viral and bacterial, including mycobacteria.<sup>3</sup> Recently *M. tuberculosis* antigens have been detected in synovial fluid of rheumatoid arthritis patients.<sup>4</sup>

Fibronectin-binding antigens are a prominent secretory protein of short term culture supernatants of *M. tuberculosis* and *M. bovis* (BCG) and is conserved within the genus *Mycobacterium*. The 30/31 kDa antigens are secreted Fibronectin-binding proteins of *M. tuberculosis* that correspond to antigens 85B and 85A of *M. bovis* BCG.<sup>5,6</sup> Fibronectin binding proteins may be important to allow the mycobacteria to avoid detection by the immune system or to facilitate interaction with the host cells.<sup>7</sup> These antigens are important targets for the human antibody response and have been successfully used as a sero diagnostic test in pulmonary TB.<sup>8,9</sup>

Abou Zeid *et al.*<sup>10</sup> demonstrated that some slow and fast, virulent and a virulent, mycobacterial species attach to fibronectin-coated surfaces, and showed that the property to bind fibronectin is conserved within the genus *Mycobacterium*. Fibronectin is known to mediate adhesion of bacteria to eucaryotic cells. It has been reported to act as an opsonin, mediating the phagocytosis of gelatin-coated particles by macrophages.<sup>11</sup>

The 30/31 kDa antigen of *M. tuberculosis* is one of the major secretory molecules and is probably routinely recognized by the host immune system in the early stage of tuberculosis infection.<sup>8</sup> In order to study the presence of this antigen in RA and TB patients, serum immune complexes were prepared from patients and controls. These were analyzed by ELISA and western blotting.

## MATERIALS AND METHODS

**RA Patients:** The study was carried out on patients with rheumatoid arthritis determined using the American Rheumatism Association criteria.<sup>12</sup>

**TB Patients:** Few confirmed positive pulmonary TB patients' serum samples were selected for the Western blotting. Patients were diagnosed on the basis of the presence of AAFB in sputum and/or following sputum culture. Data for FnBp in serum immune complexes are not shown in this paper, as the data has been reported earlier.<sup>8</sup>

**Normals:** 70 healthy normal sera were also included for the study.

**Diseased control:** Twenty nine patients' samples, other than rheumatoid arthritis, were also studied as a disease control group.

**CF8:** This monoclonal antibody (Mab) was raised against the TB- patients' immune complexes. It reacts with the 30/31 kDa fibronectin binding protein of *M. tuberculosis* and also react with the homologous BCG-85 complex.

**Fibronectin-HRP:** Human Fibronectin (Gibco) was conjugated with HRP in our laboratory and used to detect fibronectin-binding protein in the immune complexes.

**Preparation of immune complexes:** The serum immune complexes from confirmed RA, TB patients and normals were precipitated with 2% (w/v) polyethyleneglycol 6000 (PEG) overnight and washed in PEG buffer at 4°C. They were dissolved in veronal buffer.<sup>13</sup> Protein contents of immune complexes were determined by Lowry method.<sup>14</sup>

**ELISA:** Nune 96 well plates were coated with CF8 at 5 µg/ml in carbonate buffer (pH 9.6); and incubated overnight at 4°C. Excess antibodies were washed off with washing buffer (saline containing 0.05% Tween-20). The dissolved immune complexes were added in duplicate to the monoclonal antibody-coated plates and incubated at 37°C for one hour and then washed with washing buffer. A polyclonal rabbit anti-*M. tuberculosis* H37 Rv IgG-HRP conjugate (1:4000), prepared in our laboratory, was added and incubated at 37°C for 1 hour. The presence of the conjugate was detected by adding OPD (10 mg / 25 ml in citrate buffer, pH 5) with H<sub>2</sub>O<sub>2</sub>. After 30 minutes incubation the reaction was stopped by 20% H<sub>2</sub>SO<sub>4</sub>. The OD was read at 492 nm with a Lab-system Multiskan® MCC.

**Western blotting analysis:** 10ul containing 10ug of the immune complexes in protein sample buffer, containing 2-mercaptoethanol and SDS, was transferred to wells of 10% SDS-Polyacrylamide gels. Proteins were transferred to a nitrocellulose membrane for Western blot analysis. Blot was blocked with 3% Bovine serum albumin (Sigma) overnight at 4°C. Blots were stained with Fibronectin-HRP conjugate (1:1000), prepared in 5% non-fat milk, for one hour. After proper washing with PBS containing 0.1% Tween-20, bands were developed with Diamino benzedine (DAB) (50mg/100ml DAB buffer containing 25 µl of 30% H<sub>2</sub>O<sub>2</sub>). The reaction was stopped by washing blots in 0.1% (v/v) PBS-Tween 20.<sup>13</sup>

## RESULTS

Fibronectin-binding protein, the 30/31 kDa, is an important secretory protein of *M. tuberculosis* and play an important role in mycobacterial disease. We have shown

that FnBp is a characteristic feature of tuberculosis.<sup>8</sup>

The significant levels ( $p < 0.001$ ) of FnBp antigen was detected in the RA patients' serum immune complexes by using CF8 anti-FnBp m-Ab. A significant difference ( $p < 0.05$ ) was seen between late RA patients and diseased controls. However this difference was less significant between early RA patients sera and diseased controls sera (Table).

It can be clearly seen in the Figure that the immune complexes of TB patients contain FnBp in the 30/31 kDa position, as revealed by staining the blots of the separated complexes with HRP- fibronectin. Normal serum complexes give no reaction. In RA serum immune complexes FnBp was also noted but at a 37 kDa position, slightly higher than observed with the TB patients immune complexes, indicating that there are different molecular weight Fibronectin-binding protein in RA and TB patients' serum immune complexes.

In RA patients' serum immune complexes a different molecular weight, slightly higher, of 37 kDa FnBp was recognized in a western blot by using Fibronectin-HRP. This was a quite interesting finding that suggests that there are different forms of FnBp which may be involved in pathogenesis of RA. The 37 kDa FnBp could be different origin in RA, or possibly *M. tuberculosis* expresses a 37 kDa FnBp in RA. Further work is needed to prove the origin and function of 37 kDa FnBp in RA patients.

## DISCUSSION

Rheumatoid arthritis is a chronic inflammatory disease characterized by infiltration of mononuclear cells, mainly T lymphocytes, into the synovial membrane. The interaction of peripheral blood T cell with the different components of the rheumatoid synovium is mediated by cell surface proteins such as selection, fibronectin etc. T lymphocytes infiltrating the rheumatoid synovial membrane show an activated phenotype and display an increased avidity of their adherence receptor that results in enhanced interaction of these cell with both extracellular matrix protein and cellular ligand. Moreover the cells attach to extracellular matrix protein through integrin and this induces the secretion of several proteases that can contribute to the tissue damage observed in RA.

We have detected fibronectin binding protein in RA patients' serum immune complexes, which are found to play an important role in bacterial disease. We have previously shown that the 30/31 kDa FnBp is a characteristic feature of tuberculosis.<sup>8</sup> This was a quite interesting finding that suggests there are different forms of FnBp probably involved in pathogenesis of RA.

Fibronectin is present at a high concentration in rheumatoid synovium tissue and it is a chemoattractant for inflammatory cells. Fibronectin fragments play a significant role in promoting inflammation. Fibronectin is synthesized by a variety of cells including fibroblast, macrophages etc. Within joints it is found in synovial fluid, in synovial connective tissue and, in much smaller quantities, in synovial basement membrane. Fibronectin is also a chemotactic factor for fibroblasts, a function significant for repair of damaged tissue. Human fibronectin shares a common epitope with streptokinase. A cross-reactive antibody was present in plasma of 40% of patients with RA. Fibronectin and anti-fibronectin or FnBp form a complex and may play a role in the aetiology of RA.<sup>13</sup>

We have detected a 37 kDa FnBp in the serum immune complexes of RA patients, indicating the involvement of some microorganism. This could be *M. tuberculosis*, or an other mycobacterial species, or a different bacterial species. Further work is needed to prove the origin and function of the 37 kDa FnBp in RA patients.



Figure: Fibronectin-binding protein in rheumatoid arthritis, normal and TB patients' serum immune complexes.

**Tracks**

1-4: Rheumatoid arthritis patients' serum immune complexes

5-7: Normal serum immune complexes

8: Molecular weight marker

9-13: TB patients' serum immune complexes

Note: There are different molecular weight Fibronectin-binding proteins detected in rheumatoid arthritis and tuberculosis patients' serum immune complexes.

**Table**  
**Fibronectin binding protein in serum immune complexes**  
**of rheumatoid arthritis patients.**  
**Ranges, means and statistical analysis of the data.**

| Source of Samples | No. | Range         | Mean $\pm$ Standard Deviation | Standard Error | p value |        |
|-------------------|-----|---------------|-------------------------------|----------------|---------|--------|
|                   |     |               |                               |                | HC      | DC     |
| Healthy cont.     | 70  | 0.001 - 0.16  | 0.0375 $\pm$ 0.0441           | 0.0052         | -       | -      |
| Disease cont.     | 10  | 0.080 - 0.123 | 0.0960 $\pm$ 0.0132           | 0.004          | -       | -      |
| Early RA          | 12  | 0.082 - 0.494 | 0.1801 $\pm$ 0.1450           | 0.0418         | < 0.001 | > 0.05 |
| Late RA           | 24  | 0.078 - 0.258 | 0.1328 $\pm$ 0.0514           | 0.0107         | < 0.001 | < 0.05 |
| Total RA          | 36  | 0.078 - 0.494 | 0.1655 $\pm$ 0.1178           | 0.177          | -       | -      |

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